Review for Exam #2

Jericho Lawson

MATH 241

Week 10, Lecture #2 — Ch. 3-5, 6.1-6.2

Format

This class will just be used as a study hall for the most part.

- Ask questions at the beginning and during class.
- I will have a rotating pool of questions up on the screen.
- Feel free to study with those throughout class today.

Practice Question #1

Consider the following PDF for the random variable X:

$$f_X(x) = c\,x^2\,(1-x) \qquad \text{for } 0 < x < 1$$

(a) Find the value c so that the PDF is above is valid.

(b) Find the CDF for X.

(c) Find $P(0 < X < 0.5)$.

(d) Find the mean of X.

(e) Find the variance of X.

Practice Question #2

Consider the following PDF for the random variable X,

$$f_X(x) = 4x^3 \qquad \text{for } 0 < x < 1$$

Let x_0 be some value such that $F_X(x_0) = 0.20$. Find x_0.

Practice Question #3

Let $X \sim Bin(n = 3, p = \frac{2}{3})$. Let $Y = X^2$.

(a) What is the support of Y?

(b) What is the PMF of Y?

(c) Compute $P(Y = 1)$.